

ENTERPRISE GRADE AI

BioScan.AI

Biometric Face Recognition Attendance System

A full-stack, contactless biometric face recognition attendance management platform engineered for universities, colleges, and enterprise organizations. Eliminates proxy attendance with sub-second identification, single-face foreground isolation, 128-dimensional vector descriptors, and real-time Gemini Live AI voice assistance.

PROJECT SPECIFICATION & METADATA

Lead Developer:	Sahil Verma (sahilverma8085784943@gmail.com)
Domain:	Computer Vision • Biometric Security • Smart Campus Systems
Core AI Models:	Local Binary Pattern (LBP) 128-D Descriptors • Gemini Live (3.8)
Recognition Speed:	< 250 milliseconds with Euclidean distance matching
Cloud Backups:	Supabase PostgreSQL + Google Drive Spreadsheet Sync
License:	MIT Open Source License

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1. Overview

Traditional college and university attendance workflows rely heavily on paper registers, manual roll calls, or RFID smart cards. These obsolete methods suffer from critical vulnerabilities: buddy punching (students signing or swiping cards for absent peers), long bottlenecks at entrance doors, physical wear-and-tear on fingerprint sensors, and tedious manual data entry at the end of each academic semester.

BioScan.AI delivers an enterprise-grade, contactless biometric solution designed to run natively within modern client browsers. It converts video camera streams into privacy-preserving 128-dimensional mathematical vector descriptors, performs sub-second Euclidean matching, suppresses background passersby through single-face foreground isolation, and provides automated reporting to Supabase PostgreSQL and Google Drive.

2. Features

2.1 User Management & Biometric Enrollment

- **Live Webcam & File Upload Enrollment:** Capture instant webcam photos or upload high-resolution images with real-time contrast, illumination, and quality checks.
- **128-Dimensional Mathematical Vector Descriptors:** Generates 128-float normalized facial embeddings without storing invasive raw biometric templates.
- **Duplicate Roll & Vector Detection:** Automatically blocks duplicate roll numbers and flags identical facial descriptors to stop multi-profile spoofing.
- **Role-Based Access Control (RBAC):** Admin, Faculty/Staff, and Student roles with session-authenticated security passkeys for database-altering tasks.
- **Privacy-First Data Purge:** Timestamped legal consent tracking and a dedicated "Purge Biometric Data" option that removes vectors while keeping attendance history.
- **Compact Micro-Roster Preview:** Executive dashboard snippet displaying 16 enrolled student thumbnails simultaneously with status indicators.

2.2 Attendance Management & Live Kiosk Mode

- **Sub-Second Biometric Scanner HUD:** Reticle brackets, animated scanline sweep, and real-time Euclidean distance classification (threshold ≤ 0.48).
- **1-Face Focus & Background Rejection:** Dynamically isolates the primary foreground face while tagging passersby with dashed amber "[BG Ignored]" boxes.
- **Duplicate Attendance Lockout:** Prevents multiple check-ins on the same calendar day, with an authorized Admin Override modal.
- **Multi-Sensory Feedback:** High-tech sonar acquisition ping, confetti celebration particles, and clear audio chimes on verified attendance.
- **Official Daily Attendance PDF & CSV Exports:** Generates publication-ready PDF attendance sheets (via jsPDF) and downloadable CSVs.
- **Gemini Live Multimodal AI Voice Assistant:** Real-time 2-way voice kiosk assistant powered by gemini-3.8-live for hands-free queries.

3. Technologies Used

TIER	STACK / LIBRARIES	FUNCTIONAL ROLE
Frontend Framework	React 19.0.1, TypeScript 5.x, Vite 8	Component architecture, strict typing, high-speed HMR build
Styling & UI Effects	Tailwind CSS v4, Lucide Icons, Confetti	Responsive dark/light cyber aesthetic and micro-animations
Computer Vision	HTML5 Canvas 2D + LBP Texture Engine	128-dimensional normalized floating-point descriptor extraction
Matching Metric	Euclidean Vector Distance (≤ 0.48)	Nearest-neighbor vector comparison with confidence thresholding
Backend & Server	Node.js, Express, tsx, WebSockets (ws)	Full-stack server with Vite middleware & bidirectional audio streams
Database & Security	Supabase PostgreSQL, Row Level Security	Cloud database, private storage buckets, and LocalStorage cache
Multimodal AI	Google Gemini Live API (gemini-3.8-live)	16kHz microphone stream to 24kHz raw PCM real-time voice
Cloud Document Sync	Google Drive REST API + OAuth 2.0	Automated off-site cloud backups and Google Sheets sync

4. How It Works (Algorithmic Pipeline)

Step 1: Frame Acquisition
HTML5 Video streams 30fps webcam feed into an in-memory canvas context with willReadFrequently optimization.

Step 2: Spatial Clustering & Background Suppression
RGB skin-tone thresholding ($r > 60, r > g, r > b$) measures cluster centroids. If Single-Face Focus is active, peripheral clusters are marked [BG Ignored] and filtered out.

Step 3: 128-D Vector Extraction
The isolated primary face is normalized into an 8x8 grid. Local Binary Patterns (LBP) compute micro-texture and spatial luminance variances, yielding a 128-float unit-normalized vector.

Step 4: Euclidean Distance Matching
The vector is compared against all enrolled user descriptors using: $d = \sqrt{\sum (v_i - u_i)^2}$. The lowest distance is matched against the confidence threshold (0.48).

Step 5: Transaction Commit & Multi-Sensory Trigger
On successful match, duplicate same-day attendance is verified. A record is logged in Supabase / LocalStorage, accompanied by celebratory confetti and an audio chime.

5. Screenshots & UI Walkthrough

- **Executive Analytics Dashboard**

Features real-time attendance KPIs (Cohort Attendance Rate %, Present Today, Total Enrolled, Vector Quality Score), launch buttons for Kiosk Scanner, and the Mini Roster Preview.

- **Kiosk Camera Reticle HUD**

High-tech biometric targeting reticle with animated scanline sweep, real-time Euclidean distance indicator, background passerby suppression [BG Ignored], and audio feedback.

- **Biometric Enrollment & Quality Control**

Webcam snapshot or file dropzone with lighting verification, roll number uniqueness checks, interactive crop preview, and legal biometric consent checkboxes.

- **Real-Time Gemini Live AI Voice Assistant**

Bidirectional audio kiosk assistant streaming over WebSockets. Features voice visualizer orbs, persona selection (Zephyr, Puck, Kore, Fenrir), and hands-free attendance inquiries.

6. Installation and Setup Guide

Prerequisites: Node.js (v18.0.0 or higher), npm or yarn, webcam/microphone enabled browser.

Step 1: Clone Repository

```
git clone https://github.com/sahilverma8085784943/BioScan.AI-Biometric-Attendance.git
cd BioScan.AI-Biometric-Attendance
```

Step 2: Install NPM Dependencies

```
npm install
```

Step 3: Configure Environment Variables (.env)

```
GEMINI_API_KEY=your_gemini_api_key_here
VITE_SUPABASE_URL=https://your-project.supabase.co
VITE_SUPABASE_ANON_KEY=your_anon_key_here
```

Step 4: Launch Local Development Server

```
npm run dev
# Open http://localhost:3000 in Google Chrome or Microsoft Edge
```

Step 5: Production Build & Deployment

```
npm run build
npm start
```

7. Future Enhancements & Competition Roadmap

[Roadmap] Active Liveness & Anti-Spoofing Detection

Interactive blink, smile, and head-tilt challenge-response tests to prevent photo or video playback attacks from mobile screens or printed sheets.

[Roadmap] Lecture & Period-Wise Timetable Integration

Automatic class resolution mapping timestamped scans to subjects (e.g., CS301 - Data Structures), assigned faculty, and lecture hall numbers.

[Roadmap] 75% Mandatory Attendance Defaulter Engine

Automated predictive warnings calculating the exact number of classes a student can safely miss before being debarred from semester examinations.

[Roadmap] Campus GPS Geofencing & Wi-Fi BSSID Binding

Verifies that scans originate physically inside the designated lecture hall coordinates and connected to authorized university access points.

[Roadmap] Automated WhatsApp & Email Notifications

Instant check-in receipts sent to students and automated shortage warnings triggered to parents for unexcused consecutive absences.

8. Developer Info

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Academic Focus: Artificial Intelligence, Biometric Authentication, and Edge Computing.

9. License

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